4.3.1 The SVD and the four fundamental spaces

04.3.1 SVD and the Four Fundamental Subspaces

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Theorem 4.3.1.1. *Given $A \in \mathbb{C}^{m\times n}$, let $A = U_L \Sigma_{TL} V_L^H$ equal its Reduced SVD and $A = \left(\begin{array}{c|c} U_L & U_R \end{array} \right) \left(\begin{array}{c|c} \Sigma_{TL} & 0 \\ \hline 0 & 0 \end{array} \right) \left(\begin{array}{c|c} V_L & V_R \end{array} \right)^H$ its SVD. Then*

- *$\mathcal{C}(A) = \mathcal{C}(U_L)$,*
- *$\mathcal{N}(A) = \mathcal{C}(V_R)$,*
- *$\mathcal{R}(A) = \mathcal{C}(A^H) = \mathcal{C}(V_L)$, and*
- *$\mathcal{N}(A^H) = \mathcal{C}(U_R)$.*

Proof.

We prove that $\mathcal{C}(A) = \mathcal{C}(U_L)$, leaving the other parts as exercises.

Let $A = U_L \Sigma_{TL} V_L^H$ be the Reduced SVD of A. Then

- $U_L^H U_L = I$ (U_L is orthonormal),
- $V_L^H V_L = I$ (V_L is orthonormal), and
- Σ_{TL} is nonsingular because it is diagonal and the diagonal elements are all nonzero.

We will show that $\mathcal{C}(A) = \mathcal{C}(U_L)$ by showing that $\mathcal{C}(A) \subset \mathcal{C}(U_L)$ and $\mathcal{C}(U_L) \subset \mathcal{C}(A)$

- $\mathcal{C}(A) \subset \mathcal{C}(U_L)$:

  Let $z \in \mathcal{C}(A)$. Then there exists a vector $x \in \mathbb{C}^n$ such that $z = Ax$. But then $z = Ax = U_L \Sigma_{TL} V_L^H x = U_L \underbrace{\Sigma_{TL} V_L^H x}_{\widehat{x}} = U_L \widehat{x}$. Hence $z \in \mathcal{C}(U_L)$.

- $\mathcal{C}(U_L) \subset \mathcal{C}(A)$:

  Let $z \in \mathcal{C}(U_L)$. Then there exists a vector $x \in \mathbb{C}^r$ such that $z = U_L x$. But then $z = U_L x = U_L \underbrace{\Sigma_{TL} V_L^H V_L \Sigma_{TL}^{-1}}_{I} x = A \underbrace{V_L \Sigma_{TL}^{-1} x}_{\widehat{x}} = A\widehat{x}$. Hence $z \in \mathcal{C}(A)$.

We leave the other parts as exercises for the learner.

Homework 4.3.1.1. For the last theorem, prove that $\mathcal{R}(A) = \mathcal{C}(A^H) = \mathcal{C}(V_L)$.

▾ Solution

$\mathcal{R}(A) = \mathcal{C}(V_L)$:

The slickest way to do this is to recognize that if $A = U_L \Sigma_{TL} V_L^H$ is the Reduced SVD of A then $A^H = V_L \Sigma_{TL} U_L^H$ is the Reduced SVD of A^H. One can then invoke the fact that $\mathcal{C}(A) = \mathcal{C}(U_L)$ where in this case A is replaced by A^H and U_L by V_L.

Ponder This 4.3.1.2. For the last theorem, prove that $\mathcal{N}(A^H) = \mathcal{C}(U_R)$.

Homework 4.3.1.3. Given $A \in \mathbb{C}^{m\times n}$, let $A = U_L \Sigma_{TL} V_L^H$ equal its Reduced SVD and $A = \left(\begin{array}{c|c} U_L & U_R \end{array} \right) \left(\begin{array}{c|c} \Sigma_{TL} & 0 \\ \hline 0 & 0 \end{array} \right) \left(\begin{array}{c|c} V_L & V_R \end{array} \right)^H$ its SVD, and $r = \mathrm{rank}(A)$.

- ALWAYS/SOMETIMES/NEVER: $r = \mathrm{rank}(A) = \dim(\mathcal{C}(A)) = \dim(\mathcal{C}(U_L))$,
- ALWAYS/SOMETIMES/NEVER: $r = \dim(\mathcal{R}(A)) = \dim(\mathcal{C}(V_L))$,
- ALWAYS/SOMETIMES/NEVER: $n - r = \dim(\mathcal{N}(A)) = \dim(\mathcal{C}(V_R))$, and
- ALWAYS/SOMETIMES/NEVER: $m - r = \dim(\mathcal{N}(A^H)) = \dim(\mathcal{C}(U_R))$.

▾ Answer ▾ Solution

- ALWAYS: $r = \text{rank}(A) = \dim(\mathcal{C}(A)) = \dim(\mathcal{C}(U_L))$,

  The dimension of a space equals the number of vectors in a basis. A basis is any set of linearly independent vectors such that the entire set can be created by taking linear combinations of those vectors. The rank of a matrix is equal to the dimension of its columns space which is equal to the dimension of its row space.

  Now, clearly the columns of U_L are linearly independent (since they are orthonormal) and form a basis for $\mathcal{C}(U_L)$. This, together with Theorem 4.3.1.1, yields the fact that $r = \text{rank}(A) = \dim(\mathcal{C}(A)) = \dim(\mathcal{C}(U_L))$.
- ALWAYS: $r = \dim(\mathcal{R}(A)) = \dim(\mathcal{C}(V_L))$,

  There are a number of ways of reasoning this. One is a small modification of the proof that $r = \text{rank}(A) = \dim(\mathcal{C}(A)) = \dim(\mathcal{C}(U_L))$. Another is to look at A^H and to apply the last subproblem.
- ALWAYS: $n - r = \dim(\mathcal{N}(A)) = \dim(\mathcal{C}(V_R))$.

  We know that $\dim(\mathcal{N}(A)) + \dim(\mathcal{R}(A)) = n$. The answer follows directly from this and the last subproblem.
- ALWAYS: $m - r = \dim(\mathcal{N}(A^H)) = \dim(\mathcal{C}(U_R))$.

  We know that $\dim(\mathcal{N}(A^H)) + \dim(\mathcal{C}(A)) = m$. The answer follows directly from this and the first subproblem.

- ALWAYS: $r = \text{rank}(A) = \dim(\mathcal{C}(A)) = \dim(\mathcal{C}(U_L))$,
- ALWAYS: $r = \dim(\mathcal{R}(A)) = \dim(\mathcal{C}(V_L))$,
- ALWAYS: $n - r = \dim(\mathcal{N}(A)) = \dim(\mathcal{C}(V_R))$, and
- ALWAYS: $m - r = \dim(\mathcal{N}(A^H)) = \dim(\mathcal{C}(U_R))$.

Now prove it.

Homework 4.3.1.4. Given $A \in \mathbb{C}^{m\times n}$, let $A = U_L \Sigma_{TL} V_L^H$ equal its Reduced SVD and $A = \left(\begin{array}{c|c} U_L & U_R \end{array} \right) \left(\begin{array}{c|c} \Sigma_{TL} & 0 \\ \hline 0 & 0 \end{array} \right) \left(\begin{array}{c|c} V_L & V_R \end{array} \right)^H$ its SVD.

Any vector $x \in \mathbb{C}^n$ can be written as $x = x_r + x_n$ where $x_r \in \mathcal{C}(V_L)$ and $x_n \in \mathcal{C}(V_R)$.

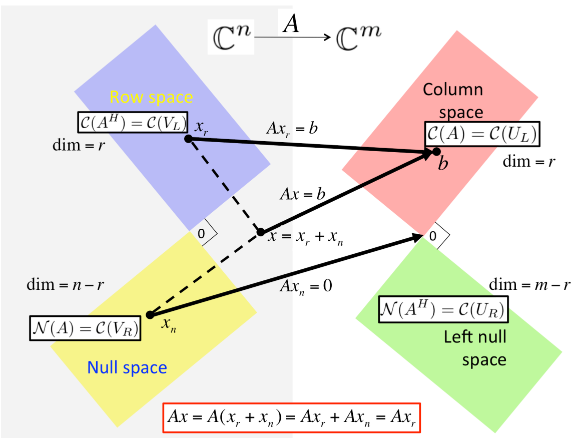
TRUE/FALSE

▼ Answer ▼ Solution

$$\begin{aligned} x &= Ix = VV^H x \\ &= \left(\begin{array}{c|c} V_L & V_R \end{array} \right) \left(\begin{array}{c|c} V_L & V_R \end{array} \right)^H x \\ &= \left(\begin{array}{c|c} V_L & V_R \end{array} \right) \begin{pmatrix} V_L^H \\ V_R^H \end{pmatrix} x \\ &= \left(\begin{array}{c|c} V_L & V_R \end{array} \right) \begin{pmatrix} V_L^H x \\ V_R^H x \end{pmatrix} \\ &= \underbrace{V_L V_L^H x}_{x_r} + \underbrace{V_R V_R^H x}_{x_n}. \end{aligned}$$

TRUE

Now prove it!



images/Chapter04/FundamentalSpacesSVD.pptx

Figure 4.3.1.2. Illustration of relationship between the SVD of matrix A and the four fundamental spaces.